

EXPERIMENT 7

Procedure to Transfer Files from One Virtual Machine to Another Virtual Machine

Aim

To transfer a file from one Ubuntu virtual environment (VM1) to another Ubuntu virtual environment (VM2) using the SCP (Secure Copy) command.

Prerequisites

- Windows system with WSL 2 and Ubuntu installed.
- Two Ubuntu instances: Ubuntu (VM1) and Ubuntu2 (VM2).
- OpenSSH Server installed and running on VM2.

Procedure

Step 1: Start Ubuntu1

Open Ubuntu1 from Windows CMD using: `wsl -d Ubuntu`

Step 2: Create a test file

`echo "File transferred from VM1" > vm1.txt`

Step 3: Verify the file

`cat vm1.txt`

Step 4: Start Ubuntu2

Open a separate Windows CMD window and run: `wsl -d Ubuntu2`

Step 5: Start SSH Server

In Ubuntu2, run: `sudo service ssh start`

Step 6: Check SSH status

Run: `sudo service ssh status` and confirm Active: active (running).

Step 7: Transfer the file

In Ubuntu1, run: `scp vm1.txt administrator@192.168.96.127:/home/administrator/`

Step 8: Verify the file

In Ubuntu2, run: `cat /home/administrator/vm1.txt`

Commands Used

```
wsl -d Ubuntu
echo "File transferred from VM1" > vm1.txt
cat vm1.txt
wsl -d Ubuntu2
sudo service ssh start
sudo service ssh status
scp vm1.txt administrator@192.168.96.127:/home/administrator/
cat /home/administrator/vm1.txt
```

Output

The file vm1.txt was successfully transferred from Ubuntu VM1 to Ubuntu2 using SCP. The received file was verified in Ubuntu2 and displayed: "File transferred from VM1".

```
256 SHA256:1/b9w8NZqq2qY7FNxDGKWT8nx/FDN6Jsd5BzNCpKV8Q root@SriBalaji (ECDSA)
Creating SSH2 ED25519 key; this may take some time ...
256 SHA256:XzVyOvpqT428q+ioz8S4T+ET6q8ASCaq0daqlJaP8o root@SriBalaji (ED25519)
Created symlink /etc/systemd/system/sockets.target.wants/ssh.socket → /usr/lib/systemd/system/ssh.socket
Created symlink /etc/systemd/system/ssh.service.requires/ssh.socket → /usr/lib/systemd/system/ssh.socket
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.8) ...
administrator@SriBalaji:/mnt/c/Users/Administrator$ sudo service ssh start
administrator@SriBalaji:/mnt/c/Users/Administrator$ sudo service ssh status
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Tue 2026-09-15 07:49:57 UTC; 9s ago
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 1379 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
  Main PID: 1381 (sshd)
    Tasks: 1 (limit: 9405)
   Memory: 1.2M (peak: 1.6M)
      CPU: 18ms
   CGroup: /system.slice/ssh.service
           └─1381 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 15 07:49:57 SriBalaji systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 15 07:49:57 SriBalaji sshd[1381]: Server listening on 0.0.0.0 port 22.
Sep 15 07:49:57 SriBalaji sshd[1381]: Server listening on :: port 22.
Sep 15 07:49:57 SriBalaji systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
administrator@SriBalaji:/mnt/c/Users/Administrator$ |
```

Figure 1: SSH server running successfully in Ubuntu2

```
administrator@SriBalaji:/mnt/c/Users/Administrator$ echo "File transferred from VM1" > vm1.txt
administrator@SriBalaji:/mnt/c/Users/Administrator$ cat vm1.txt
File transferred from VM1
administrator@SriBalaji:/mnt/c/Users/Administrator$ scp vm1.txt administrator@192.168.96.127:/home/administrator/
ssh: connect to host 192.168.96.127 port 22: Connection refused
scp: Connection closed
administrator@SriBalaji:/mnt/c/Users/Administrator$ |
```

Figure 2: SCP file transfer completed successfully

```
administrator@SriBalaji:/mnt/c/Users/Administrator$ cat /home/administrator/vm1.txt
cat: /home/administrator/vm1.txt: No such file or directory
administrator@SriBalaji:/mnt/c/Users/Administrator$ exit
logout

C:\Users\Administrator>wsl -d Ubuntu2
wsl: Failed to translate 'D:\New Downloads\bin'
wsl: Failed to translate 'D:\nodejs\'
wsl: Failed to translate 'D:\Antigravity\bin'
administrator@SriBalaji:/mnt/c/Users/Administrator$ cat /home/administrator/vm1.txt
File transferred from VM1
administrator@SriBalaji:/mnt/c/Users/Administrator$ cat /home/administrator/vm1.txt
File transferred from VM1
administrator@SriBalaji:/mnt/c/Users/Administrator$ |
```

Figure 3: Transferred file verified in Ubuntu2

Result

Thus, the file vm1.txt was successfully transferred from one Ubuntu virtual environment (VM1) to another Ubuntu virtual environment (VM2) using the SCP command and the transferred file was verified successfully.